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A nonparametric framework for linear–circular regression: Applications in environmental and biological sciences

E. Zinhom [a,*,1], M.M. Nassar [a,1], S.S. Radwan [b,1], A. Elmasry [a,1]

[a] Department of Mathematics, Faculty of Science, Ain Shams University, Cairo, Egypt
[b] Department of Mathematics, Faculty of Science Girls Section, Al-Azhar University, Cairo, Egypt

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ABSTRACT

Circular data arise in numerous scientific fields, necessitating specialized regression models to relate linear predictors to circular responses. Existing approaches often rely on restrictive parametric assumptions, limiting their flexibility in capturing complex nonlinear relationships. Here we propose a flexible nonparametric linear–circular regression framework that models circular responses as modulo transformations of latent linear variables following wrapped exponential family distributions. This leads to a finite mixture of locally weighted generalized linear models, estimated efficiently via an Expectation-Maximization algorithm. Through extensive simulations and applications to environmental and biological datasets — including droplet motion, snail movement, and wind direction — our method consistently outperforms established models in accuracy and computational efficiency. These results demonstrate the framework's adaptability and robustness in modeling complex circular data, offering a valuable tool for diverse scientific analyses involving directional outcomes.

Introduction

Circular or directional data are fundamental in numerous scientific disciplines, requiring specialized statistical and computational approaches for accurate interpretation. In biology, for example, researchers analyze animal orientation and bird flight trajectories to uncover behavioral and ecological patterns [1]. In geology, scientists investigate the alignment of the Earth's magnetic field to better understand geomagnetic dynamics [2]. With advances in imaging and sensing technologies generating vast amounts of directional information, circular statistical techniques have become increasingly valuable. A notable example is the work of Capaccioni et al. [3], who pioneered an automated imaging method to quantify the orientation of elongated mineral grains in petrographic thin sections. Orientation angles from hundreds of particles per sample were extracted and analyzed using various circular statistical tools. Angular dispersion was visualized through circular histograms and orientation frequency plots, while quantitative evaluation employed the mean resultant vector length to measure concentration around a dominant direction. To assess departures from uniformity, the Rayleigh test was applied, revealing statistically significant preferred alignments. Collectively, these analyses offered an objective and reproducible framework for detecting directional structures, providing valuable insights into geological processes such as sediment transport and rock deformation.

The integration of circular statistics within spatial statistical frameworks has gained substantial attention in recent years. Spatially referenced directional data arise in numerous applications where both the location and orientation of an observation carry essential

* Corresponding author.
E-mail address: esmailzinhom@gmail.com (E. Zinhom).
[1] All authors put equal contributions in this paper.

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information. For example, in meteorology, wind direction recorded across multiple weather stations forms a spatial field of angular measurements, while in oceanography, wave directions vary spatially across the sea surface. Similarly, in ecology, animal movement trajectories are often spatially correlated, reflecting habitat structure or environmental gradients. Modeling such data requires statistical methods that can simultaneously account for spatial dependence and the circular nature of the variables. Traditional linear spatial models are inadequate because directional data are periodic and constrained within the interval $[0, 2\pi)$. Recent advances have extended Gaussian random field models and hierarchical Bayesian frameworks to the circular domain [4–6].

The increasing availability of large-scale directional datasets in fields such as geology, biology, computer vision, and image analysis has necessitated the development of sophisticated circular statistical methods. To address the limitations of classical techniques, researchers have focused on constructing flexible circular distributions capable of modeling complex data structures [5,7]. By tailoring these distributions to specific applications, it becomes possible to capture the underlying statistical properties of angular phenomena more effectively.

A fundamental approach to generating such models is the method of wrapping, originally proposed by Lévy [8]. This process involves transforming a linear random variable Y, defined on the real line, into a circular random variable Θ via the modulo operation:

$$\Theta \equiv Y \pmod{2\pi}. \tag{1}$$

The resulting probability density function (PDF) and cumulative distribution function for $\theta \in [0, 2\pi)$ are given by the infinite series:

$$f_{\Theta}(\theta) = \sum_{k=-\infty}^{\infty} f_Y(\theta + 2k\pi) \tag{2}$$

and

$$F_{\Theta}(\theta) = \sum_{k=-\infty}^{\infty} \left[F_Y(\theta + 2k\pi) - F_Y(2k\pi)\right], \tag{3}$$

respectively.

The statistical inference and properties of wrapped distributions have been extensively explored in the literature. While classical approaches relied on the wrapped uniform, wrapped Gaussian, and wrapped Cauchy distributions [7], the framework has recently been expanded to accommodate non-symmetric and heavy-tailed baselines. This evolution has led to models such as the wrapped Laplace [9], wrapped Gamma [10], wrapped Lindley [11], wrapped modified Lindley [12], and wrapped XLindley distributions [13]. Additional contributions include the three wrapped exponential extensions proposed by Zinhom et al. [14], the wrapped Garima distribution [15], and the wrapped new XLindley distribution [16].

In parallel with these advances in circular statistics, the domain of linear distributional modeling has continued to evolve, yielding a rich collection of flexible parametric families that may serve as promising candidates for future wrapping. Notable recent developments include the generalized two-parameter discrete distribution [17], the two-parameter quadratic exponential distribution [18], the square Z-Lindley distribution [19], and a novel one-parameter family [20]. Further extensions comprise the two-parameter Beta-Exponential distribution [21], derivative models within the polynomial exponential family [22], the transmuted exponential–compound Weibull distribution for skewed data [23], the transmuted power Chris–Jerry distribution [24], and the squared XLindley distribution [25]. Collectively, these emerging linear baselines constitute a fertile foundation for the construction of new circular distributions through the wrapping mechanism defined in Eq. (2).

Although circular statistics encompass a broad range of analytical techniques, circular regression modeling specifically focuses on explaining or predicting an angular response variable using one or more explanatory factors. This approach has seen growing use across diverse fields. In biology, it is applied to model how environmental cues influence animal orientation and movement [26]. In medicine, circular regression helps interpret directional data derived from biological rhythms or medical imaging [26,27]. In environmental sciences, it assists in studying relationships between wind direction and meteorological covariates or in analyzing spatial variation in wave orientations [4,5,28]. Social scientists also use circular regression to investigate periodic behavioral trends, such as daily activity cycles or temporal voting patterns [29]. Overall, these applications highlight how circular regression extends traditional regression principles to directional domains, allowing researchers to model and interpret cyclic phenomena more accurately while maintaining the essential geometry of angular data.

Specialized regression models that can handle circular data have been developed as a result of the growing use of circular data in scientific fields. These models are divided into groups according to the types of responses and predictors that are used. In particular, there are three types of regression models: linear–circular models, which have linear predictors and circular responses [30–32]; circular–linear models, which have circular predictors and linear responses [33]; and circular–circular models, which have circular predictors and circular responses [34–40]. In this paper, variables on a real line topology are denoted by "linear", and variables on a unit circle topology by "circular". These categories provide a method for examining circular data interactions, enabling researchers to address a wide range of theoretical and practical issues in several fields.

In linear–circular regression analysis, a frequently employed modeling strategy assumes that the circular response variable Θ, given the predictor variables $X = x$, follows a von Mises distribution. The associated PDF is given by:

$$f(\Theta | g^{-1}(\beta' X), \rho) = \frac{1}{2\pi I_0(\rho)} e^{\rho \cos\left(\theta - g^{-1}(\beta' X)\right)}.$$

Here, $\rho \geq 0$ is the concentration parameter that controls the dispersion of the von Mises distribution, $I_0(\rho)$ is the modified Bessel function of the first kind and order zero, $\beta = \begin{pmatrix}\beta_0 & \beta_1 & \dots & \beta_p\end{pmatrix}'$ is a $(p+1)$-dimensional vector of regression coefficients, $X = \begin{pmatrix}1 & X_1 & \dots & X_p\end{pmatrix}'$ is a $(p+1)$-dimensional vector of predictor variables, and the term $g^{-1}(\beta' X)$ denotes the inverse of the link function [5].

Gould [41] defined the simplest form of the link function as $\mu = \beta' X$. A different formulation, $\mu = 2\pi F(X)$, representing the marginal distribution function of the predictor variables X, was later proposed by Johnson and Wehrly [42]. Later, to further broaden the framework for modeling circular regression, Fisher and Lee [36] proposed a more complicated form, $\mu = 2\tan^{-1}\left(\text{sgn}(\beta' X) \mid \beta' X \mid^{\lambda}\right)$.

Presnell et al. [43] provided compelling evidence that the three earlier methods are "challenging and generally unsatisfying". To address these issues, they introduced the Spherically Projected Multivariate Linear Model (SPML), marking a notable advancement in circular regression modeling. However, their model is built on several strong assumptions. The authors themselves acknowledge certain limitations, including the use of a simplistic covariance structure within the underlying normal model and a relatively basic representation of dispersion.

An alternative approach, proposed by Di Marzio et al. [44], employs a Non-Parametric Smoothing (NPS) technique to estimate an unknown regression function, $\mu(x)$, by minimizing the angular risk function:

$$E[1 - \cos(\Theta - \mu(x))|X = x].$$

The function that minimizes this risk is given by $\mu(x) = \arctan 2(s(x), c(x))$, where $s(x) = E(\sin(\Theta)|X = x)$ and $c(x) = E(\cos(\Theta)|X = x)$. To estimate $s(x)$ and $c(x)$, locally weighted regression was applied to the sine and cosine transformations of the observed angles, $\sin(\theta_i)$ and $\cos(\theta_i)$, respectively. The final estimate of $\mu(x)$ was then obtained by substituting replaced with the estimated values into the arctangent function: $\mu(x) = \arctan 2(s(x), c(x))$.

McCullagh and Nelder [45] introduced the framework of generalized linear models (GLMs) as a natural extension of classical linear regression, providing a more flexible approach for analyzing diverse types of data. Unlike traditional linear regression, which assumes a constant variance and a linear relationship between predictors and response, GLMs employ a link function to connect the linear predictor to the expected value of the response variable. Additionally, they allow the variance of each observation to be modeled as a function of its mean, enabling the treatment of data arising from a broad spectrum of probability distributions. Within the GLM framework, each response variable Y_i is assumed to be drawn from a probability distribution within the *exponential family*. This family is defined by a general probability density (or mass) function of the form:

$$f(y_i; \lambda_i, \phi) = \exp\left\{\frac{y_i \lambda_i - b(\lambda_i)}{a(\phi)} + c(y_i, \phi)\right\},$$

where λ_i represents the canonical (or natural) parameter, ϕ denotes the dispersion parameter, and the functions $a(\cdot)$, $b(\cdot)$, and $c(\cdot)$ specify the form of the chosen distribution (for example, normal, binomial, Poisson, or gamma).

The GLM framework connects the conditional expectation $\mu_i = E(Y_i|X_i)$ to a linear predictor through a link function $g(\cdot)$:

$$g(\mu_i) = X_i^{\top}\beta.$$

In vector notation for all n observations, the response vector is given by:

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \text{and the design matrix is} \quad X = \begin{pmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{pmatrix}.$$

Therefore, the vector of conditional means is linked to the predictors via the inverse link function as:

$$E(Y|X) = \mu = g^{-1}(X\beta).$$

The GLM framework can be extended using finite mixture models, where each component is specified as an individual GLM. This extension leads to the formulation of mixtures of generalized linear models (MGLMs) [46]. An MGLM with K components is defined as

$$f(y \mid x) = \sum_{k=1}^{K} t_k \, f_k(y \mid x; \beta_k),$$

where t_k denotes the mixing proportion of the kth component (subject to $\sum_{k=1}^{K} t_k = 1$), and $f_k(y \mid x; \beta_k)$ represents the density of the GLM associated with that component, parameterized by its coefficient vector β_k. Sikaroudi and Park [31] introduced a parametric linear–circular regression model based on a finite mixture of linear models, known as the *Angular Gaussian Mixture Model* (AGMM). Building on this foundation, our recent work [47] proposed a parametric linear–circular regression framework formulated as a finite mixture of generalized linear models, demonstrating its effectiveness in capturing heterogeneous relationships between linear predictors and circular response variables.

Huang et al. [48] introduced a finite mixture of nonparametric linear models, estimated using a modified Expectation-Maximization (EM) algorithm. They assumed that for $k = 1, 2, \dots, K$, the latent class variable K follows a discrete distribution conditional on the covariates $X = x$, given by:

$$P(K = k \mid X = x) = t_k(x),$$

where $t_k(x)$ denotes the class membership probability for component k. Conditional on $K = k$ and $X = x$, the response variable Y is assumed to follow a normal distribution with mean $m_k(x)$ and variance $\sigma_k^2(x)$, i.e.,

$$Y \mid K = k, X = x \sim \mathcal{N}\left(m_k(x), \sigma_k^2(x)\right).$$

They further assumed that the functions $t_k(\cdot)$, $m_k(\cdot)$, and $\sigma_k^2(\cdot)$ are unknown but smooth, allowing for flexible modeling of the relationship between predictors and the response.

Thus, conditioning only on $X = x$, the marginal distribution of Y is a finite mixture of normal distributions:

$$Y \mid X = x \sim \sum_{k=1}^{K} t_k(x)\, \mathcal{N}\left(\mu_k(x), \sigma_k^2(x)\right).$$

The number of components K is assumed to be fixed. This model is referred to as a nonparametric mixture of regression models because the functions $t_k(\cdot)$, $\mu_k(\cdot)$, and $\sigma_k^2(\cdot)$ are estimated non-parametrically.

Sikaroudi and Park [31] extended the method introduced by Huang et al. [48] to develop a nonparametric linear–circular regression model. This model represents a special case of the general framework presented in Section "Methodology". Therefore, we do not provide a detailed discussion of it in this paper, as it has already been thoroughly described in their original work. Statistical inference procedures and the selection of tuning parameters are addressed in Section "Model estimation". Section "Model Selection" outlines the methodology for model selection. Section "Simulation study" presents two numerical examples that test the model selection accuracy and compare the proposed method in terms of fitting and computational efficiency with the Projected Normal (PN) model [49] and the NPS approach [44]. In Section "Applications", we demonstrate the applications of the proposed models to real-world problems, including droplet motion estimation, snail movement tracking, and temporal modeling of wind direction. Finally, Section "Conclusion" concludes the paper.

Methodology

Building on the methodology proposed by Zinhom et al. [47], we develop a nonparametric framework for a linear–circular regression model. This approach utilizes a finite mixture of nonparametric GLMs to flexibly capture complex patterns in large datasets. Unlike traditional parametric approaches, this framework relaxes the assumption of a fixed functional form between predictors and the response, enabling a simply data-driven estimation of these relationships. From a practical standpoint, this flexibility reduces the manual effort required for model specification and parameter tuning. Furthermore, when coupled with modern optimization and kernel-based algorithms, the method allows for efficient and scalable estimation.

General framework

Consider a linear random variable $Y \in \mathbb{R}$ defined on the real line. This variable is transformed into a circular random variable Θ via the modulo operation $\Theta \equiv Y \pmod{2\pi}$. Since this mapping is many-to-one, the exact PDF of the circular variable, $g(\theta)$, is obtained by summing the linear density $f(y)$ over all points y that map to θ according to Eq. (2).

In the context of linear–circular regression, we assume the latent variable Y follows a distribution from the Exponential Family. Conditional on predictors $X = x$, the density of Y is defined in terms of the natural parameter $\eta(x)$ and dispersion parameter $\phi(x)$:

$$f_{Y|X}(y \mid \eta(x), \phi(x)) = \exp\left\{ \frac{y\,\eta(x) - b(\eta(x))}{a(\phi(x))} + c(y, \phi(x)) \right\}. \tag{4}$$

Here, $a(\cdot)$, $b(\cdot)$, and $c(\cdot)$ are known functions determining the specific distribution (e.g., Gaussian, Gamma, Poisson). The conditional mean and variance of the response are derived from these functions as:

$$\begin{aligned} &\mathbb{E}[Y \mid X = x] = b'(\eta(x)), \\ &\mathrm{Var}(Y \mid X = x) = b''(\eta(x))\, a(\phi(x)). \end{aligned} \tag{5}$$

Truncation and integration properties

To make the computation tractable, we restrict the infinite winding number to a finite set of integers $\mathcal{K} = \{0, \ldots, K\}$. We introduce a latent discrete winding variable Z with mixing probabilities $P(Z = k \mid X = x) = t_k(x)$, satisfying $\sum_{k\in\mathcal{K}} t_k(x) = 1$.

Given a specific winding $Z = k$ and predictors $X = x$, the conditional density of the circular variable is defined as the re-weighted linear density:

$$u(\theta \mid Z = k, X = x) = \frac{f_{Y|X}\left(\theta + 2k\pi \mid \eta(x), \phi(x)\right)}{t_k(x)}. \tag{6}$$

Marginalizing over the latent winding variable Z yields the proposed truncated model:

$$\begin{aligned} g(\theta \mid X = x) &= \sum_{k\in\mathcal{K}} t_k(x)\, u(\theta \mid Z = k, X = x) \\ &= \sum_{k\in\mathcal{K}} f_{Y|X}\left(\theta + 2k\pi \mid \eta(x), \phi(x)\right), \quad \theta \in [0, 2\pi). \end{aligned} \tag{7}$$

This representation defines a truncated wrapped exponential-family distribution, where the underlying linear model for Y induces a circular distribution for Θ.

Remark 1 (*Integration and Validity*)**.** It is crucial to observe that the function defined in Eq. (7) is an approximation of the true wrapped density. A true PDF must integrate to 1 over its domain. However, integrating Eq. (7) over $[0, 2\pi)$ yields:

$$\int_0^{2\pi} g(\theta \mid x)\, d\theta = \sum_{k\in K} \int_0^{2\pi} f_{Y|X}(\theta + 2k\pi)\, d\theta = \int_{D_K} f_{Y|X}(y)\, dy < 1, \tag{8}$$

where D_K represents the finite support covered by the chosen windings (e.g., $[0, 2\pi K]$). The result is strictly less than 1 because the probability mass of the linear distribution $f_{Y|X}(y)$ located in the "tails" (outside the truncation range K) is excluded. For the model to be valid in practice, the range K must be chosen sufficiently wide such that the missing tail probability is negligible (e.g., $<10^{-6}$).

Interpretation of mixture components

It is crucial to distinguish the role of the mixture components in the proposed wrapped framework from their role in standard finite mixture regression.

In conventional mixture models, the components $k = 1, \ldots, K$ typically represent subpopulation heterogeneity that is, distinct data-generating mechanisms (e.g., different biological groups or clusters) with independent parameters.

In contrast, the components in Eq. (7) arise from topological ambiguity rather than population heterogeneity. The index k corresponds to the "winding number" of the observation relative to the latent linear scale.

- There is only one underlying regression function, $\eta(x)$.
- The multiple components arise because a single value $\eta(x)$ on the line can map to multiple equivalent angles $\theta + 2k\pi$ on the circle.

Therefore, a high posterior probability $t_k(x) \approx 1$ does not imply the observation belongs to a specific sub-cluster, but rather that the latent response y likely lies in the interval $[2k\pi, 2(k+1)\pi]$.

Model estimation

Let $\{(\theta_1, x_1), (\theta_2, x_2), \ldots, (\theta_n, x_n)\}$ be a random sample taken from the population (Θ, X). The process for estimating our nonparametric model is explained in this section.

Expectation–Maximization algorithm (EM)

Assume $\eta(x)$ and $\phi(x)$ are smooth, nonparametric functions. The log-likelihood function for the random sample is:

$$\mathcal{L} = \sum_{i=1}^{n} \log \left[\sum_{k\in\mathcal{K}} f_{Y|X}\left(\theta_i + 2k\pi \mid \eta(x_i), \phi(x_i)\right) \right]. \tag{9}$$

We estimate these unknown functions using a kernel-based local likelihood approach, extending [48] to wrapped exponential family models. In this framework, nonparametric functions are locally approximated by kernel-weighted likelihoods, allowing for adaptive smoothing.

To ensure computational efficiency, we select a finite set of grid points $\{x^{(j)} : j = 1, 2, \ldots, J\}$. We estimate the parameters locally at these points:

$$\eta_j \approx \eta(x^{(j)}), \qquad \phi_j \approx \phi(x^{(j)}).$$

Function values at arbitrary x are subsequently obtained by interpolation (e.g., quadratic interpolation) from these grid estimates.

Let $C_h(\cdot) = h^{-1}C(\cdot/h)$ be a kernel function with bandwidth h. The localized log-likelihood centered at a specific grid point $x^{(j)}$ is given by:

$$\mathcal{L}_j = \sum_{i=1}^{n} C_h(\|x_i - x^{(j)}\|_2) \log \left[\sum_{k\in K} f_{Y|X}\left(\theta_i + 2k\pi \mid \eta_j, \phi_j\right) \right]. \tag{10}$$

Parameter estimation proceeds via an EM algorithm iteratively applied to each grid point $j = 1, \ldots, J$.

Initialize: Obtain initial estimates of the natural and dispersion parameters $\eta_j^{(0)}$ and $\phi_j^{(0)}$ for all grid points, using the initialization strategy described in Section "Parameter Initialization".

E-step: Compute the posterior responsibility $\hat{\varphi}_{i,k}^{(j)}$ that observation i (weighted by distance to grid point j) corresponds to winding number k:

$$\hat{\varphi}_{i,k}^{(j)} = \frac{f_{Y|X}\left(\theta_i + 2k\pi \mid \hat{\eta}_j, \hat{\phi}_j\right)}{\sum_{k\in\mathcal{K}} f_{Y|X}\left(\theta_i + 2k\pi \mid \hat{\eta}_j, \hat{\phi}_j\right)}. \tag{11}$$

M-step: Update the parameters (η_j, ϕ_j) for each grid point by maximizing the localized expected complete-data log-likelihood:

$$(\hat{\eta}_j, \hat{\phi}_j) = \arg\max_{\eta,\phi} \sum_{i=1}^{n} \sum_{k\in\mathcal{K}} \hat{\varphi}_{i,k}^{(j)} C_h(\|x_i - x^{(j)}\|_2) \log\left[f_{Y|X}\left(\theta_i + 2k\pi \mid \eta, \phi\right)\right]. \tag{12}$$

This optimization problem corresponds to a weighted exponential family estimation. Closed-form updates exist for certain members (e.g., Gaussian), while in general, (12) is solved numerically via Iteratively Reweighted Least Squares (IRLS). Since the local likelihood is concave, IRLS converges to a unique local maximizer [50].

Finally, the smooth mean and variance functions $\hat{\mu}(x)$ and $\hat{\sigma}^2(x)$ are recovered by interpolating the grid estimates $\hat{\mu}_j = b'(\hat{\eta}_j)$ and $\hat{\sigma}_j^2 = b''(\hat{\eta}_j)a(\hat{\phi}_j)$.

Remark 2 (*Identifiability of the Mean Function*)**.** In circular regression, the mean direction is identifiable only up to multiples of 2π. In our framework, identifiability is ensured through the initialization of the EM algorithm. A coarse "unwrapped" initial estimate (e.g., assuming small variance or using a heuristic unwrapping) anchors the model to a specific winding. Additionally, the smoothness of $\eta(x)$ discourages abrupt changes in winding across nearby covariate values, leading to a globally consistent estimation of the mean function.

Remark 3 (*Extension to Multidimensional Covariates* ($p > 1$)**.** The proposed framework extends naturally to multivariate predictors $X \in \mathbb{R}^p$ by replacing the univariate weight function with a multivariate kernel. We recommend using a Product Kernel:

$$C_{\mathbf{H}}(\mathbf{x}_i - \mathbf{x}) = \prod_{d=1}^{p} C_{h_d}(x_{id} - x_d), \tag{13}$$

where h_d represents the bandwidth for the dth dimension. This specification allows the model to adapt to varying scales across predictors.

Computational Considerations: While the mathematical formulation remains consistent, the grid-based interpolation strategy described in Section "Expectation–Maximization algorithm (EM)" suffers from the curse of dimensionality for $p > 2$ (as the grid size grows exponentially, J^p). Therefore, for higher-dimensional settings, we recommend abandoning the fixed grid in favor of Direct Evaluation, where the local likelihood parameters $(\eta(x), \phi(x))$ are estimated strictly at the target query points. For large datasets ($n > 10,000$) with high dimensions, approximation techniques such as k-d trees or binning may be employed to accelerate the neighbor search.

Parameter initialization

Standard EM algorithms are known to be sensitive to initialization. In the proposed framework, this sensitivity manifests primarily through the assignment of winding numbers K, which define the topological unwrapping of the circular response. Random initialization is insufficient because it fails to capture the continuous structure of the manifold.

To address this, we introduce a topological unwrapping procedure. By exploiting local connectivity in the joint (X, Θ) space, we estimate the initial winding numbers before entering the EM loop. This effectively linearizes the problem, placing the EM algorithm in a stable region of the likelihood surface.

Topological unwrapping via clustering

The procedure begins with the estimation of the latent labels L_i, providing a preliminary partition of the observations according to the unwrapping index. These labels are obtained by applying a clustering algorithm to the paired dataset $\{(\theta_i, x_i)\}_{i=1}^n$. In our numerical studies, we adopt the density-based clustering method introduced by Ester et al. [51].

Let I_k denote the set of observation indices assigned to the kth cluster. The number of clusters detected by the algorithm is denoted by $\hat{K}$. To align the detected clusters with the underlying topological structure, we define a mapping from clusters to integer winding indices b_k:

$$L_i = b_k, \quad \forall\, i \in I_k, \quad k \in \{1, 2, \ldots, \hat{K}\}.$$

The first cluster is assigned the reference winding index $b_1 = 0$, giving $L_i = 0$ for all $i \in I_1$. For each subsequent cluster $k \in \{2, 3, \ldots, \hat{K}\}$, the index is determined recursively. We first identify the closest previously assigned cluster $\tilde{k}$ using a Hausdorff-type distance:

$$\tilde{k} = \arg \min_{k'=1,\ldots,k-1} d_H(I_k, I_{k'}), \quad \text{where} \quad d_H(I_k, I_{k'}) = \min_{i \in I_k,\, j \in I_{k'}} \|x_i - x_j\|^2.$$

Let $(i_k, j_{\tilde{k}})$ denote the pair of points — one from the current cluster I_k and one from the previous cluster $I_{\tilde{k}}$ — that achieves this minimum. The winding index for cluster k is then defined to preserve continuity across the wrapping boundary:

$$b_k = b_{\tilde{k}} + \text{round}\left(\frac{\Theta_{j_{\tilde{k}}} - \Theta_{i_k}}{2\pi}\right). \tag{14}$$

To account for the angular wrapping effect, the unwrapped angles for observations in cluster k are computed as:

$$Y_{i \in I_k} = \Theta_{i \in I_k} + 2\pi b_k.$$

Initial parameter estimation

Using the unwrapped responses $y^{(0)}$, we construct initial estimates of the global functions $\eta(x)$ and $\phi(x)$ to "warm start" the EM algorithm.

We employ a B-spline basis representation. Let $\mathbf{B}$ be the $n \times J$ design matrix of B-spline basis functions and $\mathbf{y}^{(0)}$ be the vector of unwrapped responses. We estimate the coefficients $\hat{\boldsymbol{\beta}}$ via ridge-regularized least squares:

$$\hat{\boldsymbol{\beta}} = (\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{I})^{-1} \mathbf{B}^\top \mathbf{y}^{(0)}. \tag{15}$$

To ensure numerical stability during initialization, a ridge penalty λ is introduced to mitigate possible ill-conditioning of $\mathbf{B}^T\mathbf{B}$, which can occur when clusters are small or basis functions are strongly correlated. We set $\lambda = 0.1$, which provides sufficient regularization to guarantee invertibility without materially biasing the initial estimates. Empirical evaluations confirm that the initialization procedure is insensitive to moderate variations in λ.

This yields an initial mean function estimate $\hat{\mu}^{(0)}(x_i) = (\mathbf{B}\hat{\boldsymbol{\beta}})_i$. The initial natural parameter $\hat{\eta}^{(0)}(x)$ is obtained via the canonical link function:

$$\hat{\eta}^{(0)}(x) = (b')^{-1}(\hat{\mu}^{(0)}(x)).$$

An initial constant dispersion $\hat{\phi}^{(0)}$ is estimated from the variance of the residuals $r_i = y_i^{(0)} - \hat{\mu}^{(0)}(x_i)$. These values $(\hat{\eta}^{(0)}, \hat{\phi}^{(0)})$ are passed directly to the first E-step of the algorithm described in Section "Expectation–Maximization algorithm (EM)".

Circular prediction and boundary continuity

A standard challenge in circular regression is the "boundary artifact" problem, where conventional smoothing techniques fail near the cut-point (typically $0/2\pi$). For instance, calculating the arithmetic mean of angles near 0 and 2π often results in an erroneous estimate near π.

Our framework resolves this by performing all estimation on the unwrapped manifold. Since the latent variable Y is defined on the real line $\mathbb{R}$, the kernel smoothing operation:

$$\hat{\mu}(x) = \frac{\sum_{i=1}^{n} C_h(x - x_i)\, y_i}{\sum_{i=1}^{n} C_h(x - x_i)} \tag{16}$$

is computed using the unwrapped responses $y_i = \theta_i + 2k_i\pi$. In this linear space, the boundary at $0/2\pi$ is merely a continuous point on the line (e.g., the transition between winding k and $k+1$).

The final circular prediction is obtained by projecting this linear estimate back onto the unit circle:

$$\hat{\theta}(x) = \hat{\mu}(x) \pmod{2\pi}. \tag{17}$$

This predict-then-wrap approach guarantees that the resulting circular mean is continuous and free from the edge-bias artifacts that plague standard circular kernel regression.

Theoretical properties

While a full asymptotic characterization of the hybrid estimator is complex due to the topological initialization, we outline the theoretical guarantees inherited from the underlying local likelihood and EM frameworks.

Convergence of the estimation algorithm

The parameter estimation relies on a localized EM algorithm. A fundamental property of the EM algorithm is the monotonic ascent of the observed-data log-likelihood function $\mathcal{L}(\theta)$ at each iteration [52]. Since the localized likelihood function in Eq. (10) is bounded above (for kernels with compact support and bounded densities), the sequence of likelihood values converges. Following Wu [53], provided the likelihood surface is continuous, the algorithm is guaranteed to converge to a stationary point (local maximum).

Consistency of the estimator

Conditional on the correct identification of the winding numbers, the problem reduces to estimating the smooth parameters $\eta(x)$ and $\phi(x)$ of an exponential family distribution via kernel-weighted likelihood. Standard results from nonparametric smoothing theory apply here. Specifically, following Fan and Gijbels [54], if the bandwidth h satisfies $h \to 0$ and $nh \to \infty$ as $n \to \infty$, and assuming the underlying functions η and ϕ are twice continuously differentiable, the local likelihood estimator is consistent:

$$\hat{\eta}(x) \xrightarrow{p} \eta(x) \quad \text{and} \quad \hat{\phi}(x) \xrightarrow{p} \phi(x). \tag{18}$$

This implies that as sample size increases, the estimated latent curve converges in probability to the true unwrapped mean function.

Model selection

Model selection for the proposed framework involves determining two key hyperparameters: the truncation limit (winding number) K, and the kernel bandwidth h. While bandwidth selection can be effectively handled via data-driven approaches like cross-validation (CV), selecting K requires balancing computational fidelity with the topological structure of the data.

Selection of the truncation limit K

In the context of wrapped distributions, the parameter K does not represent the number of distinct clusters (as in finite mixture models), but rather the extent of the wrapping structure considered. Determining the optimal K is crucial: selecting too small a K fails to capture the full support of the latent distribution (bias), while an excessively large K increases computational cost without improving fit.

We adopt a likelihood-based information criterion approach. The general form is given by:

$$IC = -2\mathcal{L} + \alpha \times df, \tag{19}$$

where $\mathcal{L}$ denotes the maximized log-likelihood, and df represents the effective degrees of freedom. We consider the Akaike Information Criterion (AIC, $\alpha = 2$) and the Bayesian Information Criterion (BIC, $\alpha = \log(n)$) as practical likelihood-based criteria.

For the proposed non-parametric wrapped model, the degrees of freedom are primarily determined, to a first-order approximation, by the smoothness of the latent functions $\eta(x)$ and $\phi(x)$, rather than the number of windings. Following Huang et al. [48], the effective degrees of freedom for a kernel-based estimator is approximated by:

$$df \approx 2 \times \nu_C h^{-1} |\Omega| \left(C(0) - \frac{1}{2} \int C^2(t) dt \right), \tag{20}$$

where the factor 2 accounts for the two latent functions (μ and σ^2), h is the bandwidth, $|\Omega|$ is the range of the covariate space, and ν_C is a kernel-dependent constant defined as:

$$\nu_C = \frac{C(0) - \frac{1}{2} \int C^2(t) dt}{\int \left\{ C(t) - \frac{1}{2} C * C(t) \right\}^2 dt}. \tag{21}$$

Here, $C * C(t)$ denotes the convolution of the kernel function with itself.

Remark 4 (*Asymmetric Bias-Variance Trade-Off*)**.** The selection of K exhibits an asymmetric sensitivity. Under-estimation ($K < K_{\text{true}}$) introduces severe bias, as the model cannot account for data points on the outer windings, effectively "cutting off" the tails of the distribution. Over-estimation ($K > K_{\text{true}}$) is benign; the EM algorithm implicitly assigns negligible probability mass to the superfluous windings ($\hat{r}_{ik} \approx 0$). Therefore, we recommend prioritizing a K that minimizes BIC but verifying that the estimated posterior probabilities for the boundary windings ($k = \pm K$) are near zero.

Diagnostic validation of truncation

While information criteria provide a global selection metric, we recommend a post-estimation diagnostic check to ensure the validity of the approximation. Since the model relies on truncating the infinite sum of windings, adequate coverage of the latent distribution tails is essential.

We propose the Boundary Mass Diagnostic. Let $\hat{\varphi}_{i,k}$ be the estimated posterior probability that observation i belongs to winding k. We compute the total probability mass assigned to the boundary windings $\mathcal{K}_{bd} = \{0, K\}$:

$$M_{boundary} = \frac{1}{n} \sum_{i=1}^{n} \sum_{k \in \mathcal{K}_{bd}} \hat{\varphi}_{i,k}. \tag{22}$$

If $M_{boundary} > \epsilon$ (where ϵ is a pre-defined tolerance, e.g., 0.01), it suggests that the latent linear distribution extends beyond the current truncation window. In such cases, the range K should be expanded (e.g., $K \leftarrow K + 1$) and the model re-estimated until the boundary mass is negligible. This ensures that the truncation error is statistically insignificant relative to the estimation noise.

Bandwidth selection

Given a selected K, we propose a J-fold Cross-Validation (CV) procedure to determine the optimal bandwidth h.

Let $\mathcal{D}$ denote the full dataset partitioned into J disjoint folds. For each fold $j = 1, \ldots, J$, we train the model on the remaining data to obtain estimates $\hat{\mu}^{(-j)}(\cdot)$ and $\hat{\sigma}^{2(-j)}(\cdot)$.

We evaluate the predictive performance on the test fold T_j using the circular absolute error. For each observation $(x_i, \theta_i) \in T_j$, the predicted angle is given by projecting the estimated latent mean onto the circle:

$$\hat{\theta}_i = \hat{\mu}^{(-j)}(x_i) \pmod{2\pi}. \tag{23}$$

The Cross-Validation score is defined as:

$$CV(h) = \sum_{j=1}^{J} \sum_{i \in T_j} \left| \sin\left(\frac{\theta_i - \hat{\theta}_i}{2} \right) \right|. \tag{24}$$

We select the bandwidth h that minimizes this score.

Joint selection strategy

Since K and h can interact — specifically, oversmoothing (large h) can artificially merge distinct winding layers — we recommend a two-step strategy:

1. **Grid Search for K:** Fix a small pilot bandwidth and select K by minimizing the BIC. This ensures the topological structure is large enough to contain the data.
2. **CV for h:** Fix the selected K and refine the bandwidth h by minimizing the CV score in Eq. (24).

A complete summary of the estimation procedure, including initialization, model selection, and the EM algorithm, is provided in Algorithm 1.

Algorithm 1 Hybrid Nonparametric Estimation Framework

Input: Dataset $\{(\theta_i, x_i)\}_{i=1}^{n}$, Grid points $\{x^{(j)}\}_{j=1}^{J}$, Candidates $\mathcal{K}, \mathcal{H}$.
Output: Estimated functions $\hat{\mu}(x)$ and $\hat{\sigma}^2(x)$.

Step 1: Initialization
Apply DBSCAN to $\{(\theta_i, x_i)\}$ (using cylindrical distance) to identify clusters and winding numbers c_m.
Unwrap responses: $y_i = \theta_i + 2\pi c_m$.
Estimate initial global functions $\hat{\boldsymbol{\eta}}^{(0)}$ and $\hat{\boldsymbol{\phi}}^{(0)}$ via B-spline ridge regression on y_i.
Step 2: Model Selection
for all $K \in \mathcal{K}$ **do**
  for all $h \in \mathcal{H}$ **do**
    Run EM Algorithm (Step 3) to convergence.
    Calculate BIC (for K).
  end for
end for
Set optimal parameters: $K^* \leftarrow \arg\min(\text{BIC})$
for all $h \in \mathcal{H}$ **do**
  Cross-Validation error (for h).
end for
Set optimal parameters: $h^* \leftarrow \arg\min(\text{Cross-Validation error})$.
Step 3: EM Estimation (at optimal K^*, h^*)
while change $> \epsilon$ **do**
  E-step: Compute posterior responsibilities $\hat{\varphi}_{i,k}$ for each winding k Equation (11).
  M-step (Exp. Family Parameters): Update $\hat{\eta}(x), \hat{\phi}(x)$ locally at grid points Equation (12):
  if Distribution is Gaussian **then**
    Solve via **Weighted Least Squares** (Closed-form).
  else
    Solve via **IRLS**.
  end if
end while
Step 4: Prediction & Wrapping
Interpolate $\hat{\mu}(x)$ from grid points to target locations.
Compute final circular prediction: $\hat{\theta}(x) = \hat{\mu}(x) \pmod{2\pi}$.

Bandwidth and topological interaction

The interaction between the smoothing bandwidth h and the mixture structure K is a defining characteristic of the wrapped framework. We explicitly analyzed this dependency in our simulation studies (Tables 1 and 4).
Effect of Oversmoothing: We observe that excessive smoothing ($h \gg h_{opt}$) can lead to "topological collapsing". As shown in Table 1 (True $K = 2$), increasing the bandwidth to $h = 0.3$ resulted in a significant increase in the selection of $K = 1$ (77 cases out of 100). Theoretical justification suggests that as $h \to \infty$, the estimator $\hat{\mu}(x)$ approaches a global linear fit, which lacks the flexibility to represent multiple winding sheets. Consequently, the model forces distinct layers to merge, inducing a structural bias that BIC correctly penalizes.
Effect of Undersmoothing: Conversely, undersmoothing ($h \to 0$) generally preserves the separation between winding components but increases the variance of the estimate within each winding. As seen in Table 4 (True $K = 3$), the selection of K remained robust at low bandwidths ($h = 0.01$), indicating that component separation is less sensitive to variance than to bias. This asymmetry supports our strategy of prioritizing K selection (to fix topology) before refining h via cross-validation.

Simulation study

This section presents two numerical examples illustrating the effectiveness of the proposed framework. We consider Gamma and Laplace distributions for the latent response variable Y, each paired with a distribution-specific kernel: a Gaussian kernel for the Gamma case and a Laplace kernel for the Laplace case. Quadratic interpolation is applied in both models, as described earlier. The proposed model is compared with two established approaches: the PN model [49] and a NPS model [44]. For the PN model, several mean structures (linear, quadratic, cubic) are evaluated following the original source, but only the best-performing result for each example is reported.

Model performance is assessed using three complementary metrics to capture different aspects of prediction accuracy: Mean Circular Error (MCE), the Circular Correlation Coefficient (r_{circ}), and Cosine Similarity (CS).

First, the MCE measures the average angular discrepancy between the predicted and observed values. The angular error for a single observation is defined as:

$$\left|\sin\left(\frac{\theta_i - \hat{\theta}_i}{2}\right)\right|,$$

which lies within $[0, 1]$, making it bounded and interpretable. The MCE is computed across all n test observations as:

$$\text{MCE} = \frac{1}{n}\sum_{i=1}^{n}\left|\sin\left(\frac{\theta_i - \hat{\theta}_i}{2}\right)\right|, \tag{25}$$

where θ_i is the ground truth angle at the ith test location and $\hat{\theta}_i$ is the predicted value. Lower MCE values indicate better performance.

To evaluate the strength of the linear relationship between the observed and predicted angles, we employ the r_{circ}, defined as:

$$r_{circ} = \frac{\sum_{i=1}^{n}\sin(\theta_i - \bar{\theta})\sin(\hat{\theta}_i - \bar{\hat{\theta}})}{\sqrt{\sum_{i=1}^{n}\sin^2(\theta_i - \bar{\theta})\sum_{i=1}^{n}\sin^2(\hat{\theta}_i - \bar{\hat{\theta}})}}, \tag{26}$$

where $\bar{\theta}$ and $\bar{\hat{\theta}}$ denote the sample circular means of the true and predicted angles, respectively. Values of r_{circ} close to 1 indicate a strong positive correlation.

Finally, we assess the closeness of predictions on the unit circle using CS, calculated as:

$$\text{CS} = \frac{1}{n}\sum_{i=1}^{n}\cos(\theta_i - \hat{\theta}_i). \tag{27}$$

This metric ranges from -1 to 1, where a value of 1 implies a perfect match between the predicted and observed directions.

While estimation accuracy is the primary evaluation criterion, computational efficiency is also considered, particularly for large-scale or time-sensitive applications. Each model is run 30 times per configuration, and the average runtime is recorded. For the proposed and NPS models, a five-fold cross-validation is used to select the optimal kernel bandwidth. The PN model uses 30,000 iterations of Gibbs sampling, discarding the first 10,000 as burn-in. All experiments are conducted on a system with an Intel Core i3-10110U CPU and 12 GB RAM.

Parameter estimation for the PN model is performed using the `bpnreg` R package, and the NPS model is implemented with the `NPCirc` R package. The datasets are synthetically generated based on scenarios from [31,32], using the `circular` package in R.

Remark 5 (*Choice of Kernel Function*)**.** The choice of the smoothing kernel $C(\cdot)$ is guided by practical considerations of localization and robustness rather than asymptotic requirements. For the Gamma simulation scenarios, we employ the standard Gaussian kernel ($C(u) \propto e^{-u^2/2}$), which provides smooth weighting and strong localization. For the Laplace simulation scenarios, we use the Laplace kernel ($C(u) \propto e^{-|u|}$), which induces sharper spatial decay and has been found to offer improved robustness in the presence of heavy-tailed noise. Importantly, the asymptotic properties of the proposed estimator are driven primarily by the bandwidth choice, and the results remain valid for a broad class of kernel functions [55].

Example 1: Truncated Wrapped Laplace Model (TWLM)

The datasets in this example are generated as follows. The covariate X is sampled from a uniform distribution on $[0, 1]$. For each observed x, the conditional distribution $\Theta \mid X = x$ follows a von Mises distribution with mean direction

$$\mu(\Theta \mid X = x) = 0.2 + \tan^{-1}(10x),$$

and concentration parameter ρ. Smaller values of ρ correspond to greater dispersion around the mean direction. The latent variable Y follows a Laplace distribution.

Table 1
Frequency of selected number of components K in Example 1 across 1000 simulations for varying bandwidths h and sample sizes n, with fixed $\rho = 8$.

Sample size	Number of components	$h = 0.05$	$h = 0.1$	$h = 0.15$	$h = 0.2$	$h = 0.3$	$\min_h BIC_h$
$n = 200$	$K = 1$	5	0	0	8	77	7
	$K = 2$	994	1000	1000	991	923	992
	$K = 3$	1	0	0	1	0	1
	$K = 4$	0	0	0	0	0	0
	$K = 5$	0	0	0	0	0	0
$n = 400$	$K = 1$	0	0	0	0	16	0
	$K = 2$	1000	1000	1000	1000	984	1000
	$K = 3$	0	0	0	0	0	0
	$K = 4$	0	0	0	0	0	0
	$K = 5$	0	0	0	0	0	0
$n = 800$	$K = 1$	0	0	0	0	1	0
	$K = 2$	1000	1000	1000	1000	999	1000
	$K = 3$	0	0	0	0	0	0
	$K = 4$	0	0	0	0	0	0
	$K = 5$	0	0	0	0	0	0

Table 2
Comparison MCE, CS, and r_{circ} of the estimated mean functions for the three compared approaches in Example 1 over varying ρ values.

ρ	Metric								
	MCE			CS			r_{circ}		
	TWLM	PN	NPS	TWLM	PN	NPS	TWLM	PN	NPS
2	0.0859	0.1211	0.0949	0.9670	0.9605	0.9667	0.9876	0.9924	0.9930
4	0.0446	0.0494	0.0497	0.9928	0.9916	0.9913	0.9983	0.9993	0.9981
6	0.0258	0.0336	0.0489	0.9975	0.9955	0.9959	0.9993	0.9991	0.9985
8	0.0251	0.0289	0.0506	0.9974	0.9961	0.9966	0.9992	0.9993	0.9987

Model selection

We evaluate the effectiveness of the BIC-based information criterion in Eq. (19) for selecting the number of mixture components. For each simulated dataset, nonparametric mixture regression models with $K \in \{1, 2, 3, 4, 5\}$ components are fitted using bandwidths $h \in \{0.05, 0.1, 0.15, 0.2, 0.3\}$, chosen to span a range from undersmoothing to oversmoothing.

For $\rho = 8$ and each fixed bandwidth h, Table 1 reports the frequency with which each K is selected across 1000 simulations. Across all sample sizes, the BIC-based procedure overwhelmingly identified the true $K = 2$ model for most bandwidths, with selection frequencies exceeding 99% in nearly all cases. For $n = 200$, slight deviations occurred at $h = 0.05$ and $h = 0.2$ (i.e. for under and over smoothing), where a small proportion of runs selected $K = 1$. As the sample size increased to $n = 400$ and $n = 800$, the procedure achieved perfect or near-perfect accuracy, even for larger bandwidths.

When K was chosen by jointly minimizing the BIC over all (K, h) combinations, the final column of Table 1 shows that the true $K = 2$ model was correctly identified in 99.2%, 100%, and 100% of cases for $n = 200$, 400, and 800, respectively. These results confirm the robustness and high effectiveness of the BIC-based criterion under the given simulation settings.

Model fitting

Datasets of size $n = 200$ are generated for concentration parameter values $\rho \in \{2, 4, 6, 8\}$. Table 2 reports the MCE, CS, and r_{circ} of the estimated mean functions for three approaches: the proposed TWLM method, the PN model, and NPS.

To ensure the reliability of the TWLM estimates, the EM algorithm employed a strict stopping criterion, terminating only when the change in parameter estimates fell below 10^{-9}. Fig. 1 illustrates the training history, showing the monotonic increase and stabilization of the global log-likelihood, which confirms the convergence of the estimated parameters.

The proposed TWLM demonstrates superior accuracy and robustness, achieving the best performance in all scenarios. The advantage is most pronounced at smaller ρ values (e.g., $\rho = 2$), indicating greater stability under high variability; here, TWLM achieves the lowest MCE (0.0859) and the highest Cosine Similarity (0.9670). The PN model, while competitive, generally yields larger errors, suggesting that its parametric structure may limit flexibility. The high r_{circ} values for TWLM (consistently > 0.99) further confirm that the non-parametric approach effectively captures the underlying directional trend. NPS generally performs weakest, particularly in low-concentration settings, where noise hinders the accurate recovery of the underlying structure.

Fig. 2 compares the estimated mean functions from all methods. TWLM closely tracks the true function across the entire range of X, while the other methods exhibit noticeable deviations, especially in regions with rapid directional change.

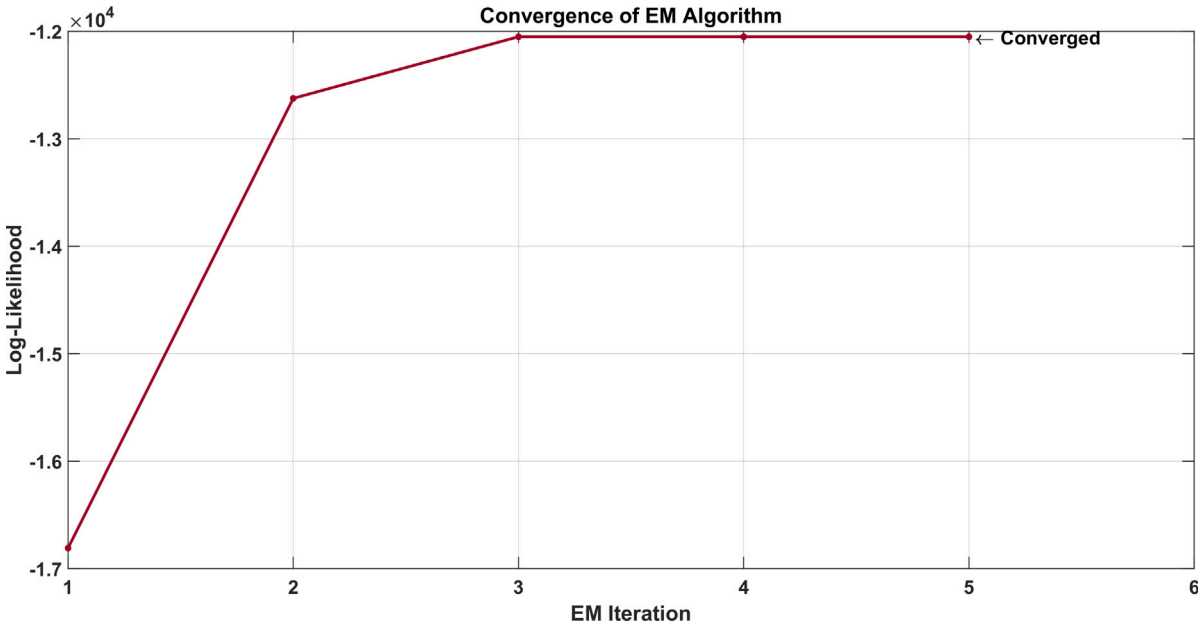


Fig. 1. Convergence of the EM algorithm for the proposed TWLM model in Example 1 when $\rho = 6$.

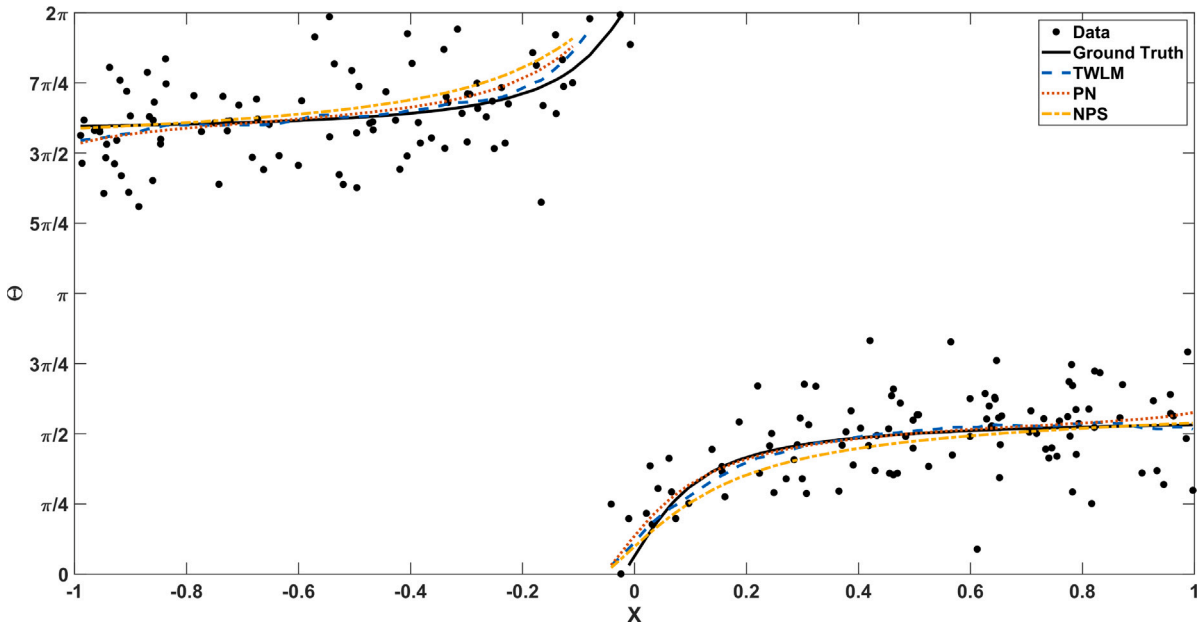


Fig. 2. Estimated mean functions for the three models and ground truth in Example 1 when $\rho = 6$.

Computational aspect

Table 3 summarizes the average computation times (in seconds) for the three methods across different ρ values, based on a sample size of $n = 200$. It is important to note that the reported times for the proposed TWLM method include the entire model selection process, specifically the 5-fold cross-validation used for bandwidth selection.

After confirming runtime normality, a paired t-test comparing TWLM and NPS yields a p-value of 0.184, indicating no statistically significant difference. However, the PN model requires substantially more time due to the computational cost of Gibbs sampling.

Table 3
Average computation times (in seconds) for the three methods in Example 1 across different ρ values (sample size $n = 200$).

ρ	TWLM	PN	NPS
2	1.2845	18.5170	0.6726
4	0.9711	18.4727	0.6363
6	0.3407	18.5174	0.3406
8	0.3286	18.1655	0.2911

Table 4
Frequency of selected number of components K in Example 2 across 1000 simulations for varying bandwidths h and sample sizes n, with fixed $\rho = 8$.

Sample size	Number of components	$h = 0.01$	$h = 0.05$	$h = 0.1$	$h = 0.15$	$h = 0.2$	$\min_h BIC_h$
$n = 200$	$K = 1$	0	0	0	0	0	0
	$K = 2$	0	0	0	0	1	1
	$K = 3$	998	1000	1000	1000	999	997
	$K = 4$	2	0	0	0	0	2
	$K = 5$	0	0	0	0	0	0
$n = 400$	$K = 1$	1	0	0	0	0	0
	$K = 2$	0	0	2	2	0	1
	$K = 3$	999	1000	998	998	1000	999
	$K = 4$	0	0	0	0	0	0
	$K = 5$	0	0	0	0	0	0
$n = 800$	$K = 1$	0	0	0	0	0	0
	$K = 2$	1	3	0	2	2	3
	$K = 3$	999	997	1000	998	998	997
	$K = 4$	0	0	0	0	0	0
	$K = 5$	0	0	0	0	0	0

Example 2: Truncated Wrapped Gamma Model (TWGM)

In this example, the data are generated following the same procedure described in Example 1, except that the conditional distribution $\Theta \mid X = x$ has a mean direction given by

$$\mu(\Theta \mid X = x) = 7(x - 0.5) + 1.$$

The latent variable Y is assumed to follow a Gamma distribution.

Model selection

For $\rho = 8$, Table 4 shows the frequency with which each number of components K was selected across 1000 simulated datasets for various fixed bandwidths h. Across all sample sizes, the BIC-based procedure consistently selected the true three-component model ($K = 3$) with extremely high accuracy, typically exceeding 99.7%.

For $n = 200$, $K = 3$ was selected in almost all simulations, with only rare misclassifications into $K = 4$ or $K = 2$ (the latter occurring solely at $h = 0.2$). At $n = 400$ and $n = 800$, the results were similarly strong, with perfect or near-perfect selection across all bandwidths. Small deviations—mainly to $K = 2$—were observed at a few bandwidth values, but the error rates remained below 0.3%.

The final column of Table 4 presents the results from joint optimization over (K, h) via $\min_h BIC_h$. This approach achieved correct selection rates of 99.7%, 99.9%, and 99.7% for $n = 200$, 400, and 800, respectively, confirming the robustness and reliability of the BIC-based criterion across different smoothing levels and sample sizes.

Model performance

We generate datasets of size $n = 400$ for concentration parameter values $\rho \in \{2, 4, 6, 8\}$ to evaluate the model under varying noise levels. Table 5 presents the comparison of the estimated mean functions for three methods: TWGM, the PN model, and NPS. The results are reported using MCE, CS, and r_{circ}.

Convergence of the EM algorithm was assured through a strict tolerance (10^{-9}) on parameter updates. Fig. 3 illustrates the resulting training trajectory, where the log-likelihood exhibits the expected monotonic rise before stabilizing, demonstrating successful parameter estimation for the TWGM.

The proposed TWGM consistently produced very low errors, with MCE values ranging from 0.0259 to 0.0388 and higher CS across all ρ values. The NPS method achieved similar accuracy with higher r_{circ} across all ρ except when $\rho = 4$, yielding results comparable to the proposed framework.

In contrast, the PN model yielded substantially larger errors, with MCE values exceeding 0.56 for all ρ values, indicating considerably lower estimation accuracy compared to the other two methods. Both TWGM and NPS maintained stable performance

Table 5
Comparison MCE, CS, and r_{circ} of the estimated mean functions of the three compared approaches in Example 2 over a range of ρ values.

ρ	Metric								
	MCE			CS			r_{circ}		
	TWGM	PN	NPS	TWGM	PN	NPS	TWGM	PN	NPS
2	0.0370	0.5987	0.0471	0.9945	0.0835	0.9935	0.9842	0.0645	0.9914
4	0.0383	0.5778	0.0340	0.9950	0.1456	0.9963	0.9938	0.1272	0.9925
6	0.0360	0.5677	0.0404	0.9953	0.1755	0.9946	0.9613	0.1732	0.9818
8	0.0259	0.5628	0.0299	0.9976	0.1719	0.9956	0.9508	0.1699	0.9781

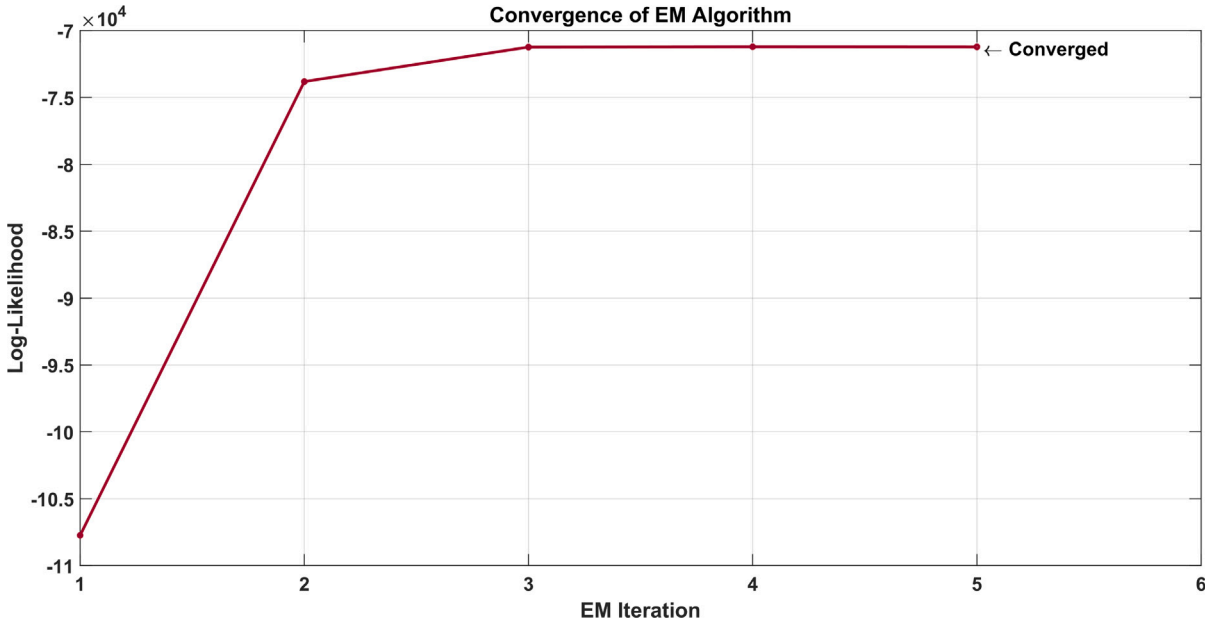


Fig. 3. Convergence of the EM algorithm for the proposed TWGM model in Example 2 when $\rho = 2$.

Table 6
Average computation time (in seconds) of the three compared methods in Example 2 across different values of ρ (sample size $n = 400$).

ρ	TWGM	PN	NPS
2	0.4901	24.7257	1.3654
4	0.3442	25.5924	1.0039
6	0.3311	24.6857	0.7748
8	0.3345	24.6437	0.8414

as ρ increased, demonstrating robustness to changes in concentration. The PN method showed only slight improvement with higher ρ, highlighting its limitation in capturing the topological winding structure of the data. Fig. 4 provides a visual comparison for the case $\rho = 2$, illustrating the estimated models alongside the data and ground truth mean.

Computational aspect

Table 6 presents the average computation times for the three methods applied in Example 2, across different values of the noise parameter ρ and sample size $n = 400$. A paired t-test was again used to compare TWGM and NPS after validating the normality assumption. The resulting p-value of 0.007 indicates a statistically significant difference in runtime between the two methods, with TWGM exhibiting notably faster convergence.

Similar to Example 1, the PN model consistently demonstrated the highest computation time, with average runtimes exceeding 24 s across all ρ values.

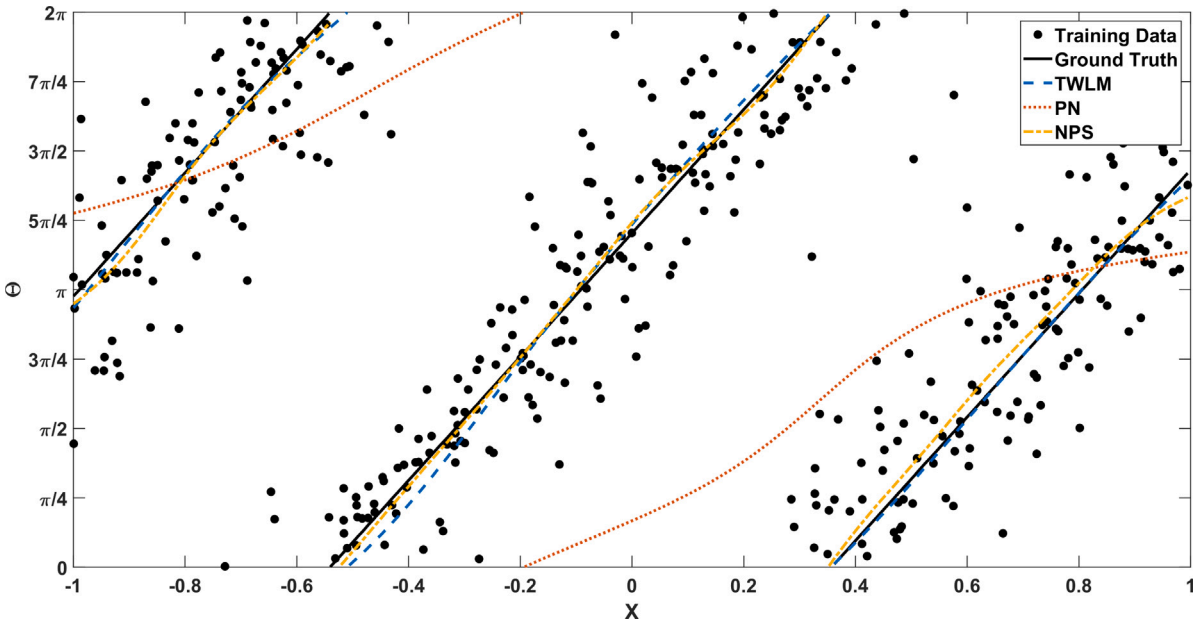


Fig. 4. The estimated mean functions for three models and ground truth in Example 2 when $\rho = 2$.

Table 7
Performance metrics and computation time (in seconds) for the droplet motion prediction using the four comparative approaches.

Metric	TWLM	TWGM	PN	NPS
MCE	0.0064	0.0055	0.0076	0.0072
CS	0.9999	0.9999	0.9998	0.9998
r_{circ}	0.9864	0.9873	0.9772	0.9797
Time (s)	0.0614	0.0183	0.7018	0.0688

Applications

In this section, we apply the proposed framework, utilizing the two previously introduced exponential family distributions (Gamma and Laplace), to three real-world case studies. To ensure robust estimation of local curvature and minimize boundary bias, we implement the M-step using local quadratic regression (a hybrid of kernel weighting and polynomial fitting) rather than standard local constant smoothing. We compare the predictive performance of this approach against two established benchmarks: the PN model and the standard NPS model.

Droplet motion analysis

The first real-world example draws upon the experimental study by Abbott [56], which investigated the motion of a droplet moving at variable velocity through a viscous medium. While the original work established theoretical laws supported by extensive empirical measurements, our analysis focuses on a specific subset of the data: 12 observations recording the angle of lead of the electromotive force ahead of a falling droplet (measured in degrees) against varying wavelengths (in centimeters).

Table 7 summarizes the predictive performance using MCE, CS, and r_{circ}, alongside the total computation time. Fig. 5 visualizes the estimated mean functions relative to the observed data points.

Given the small sample size and clean functional relationship of this dataset, all four methods achieved high accuracy, with Cosine Similarity scores nearing 1.0. However, the proposed TWGM framework demonstrated the best overall performance, achieving the lowest MCE (0.0055) and the highest circular correlation (0.9873).

The proposed TWLM and the standard NPS method also performed well, yielding MCE values of 0.0064 and 0.0072, respectively. In terms of computational efficiency, TWGM was the fastest (0.0183 s), followed closely by TWLM (0.0614 s).

In contrast, the PN model, while accurate (MCE = 0.0076), was significantly less efficient. Its runtime of 0.7018 s—while fast in absolute terms—was approximately 38 times slower than the proposed TWGM method, highlighting the overhead of its sampling-based inference even on small datasets.

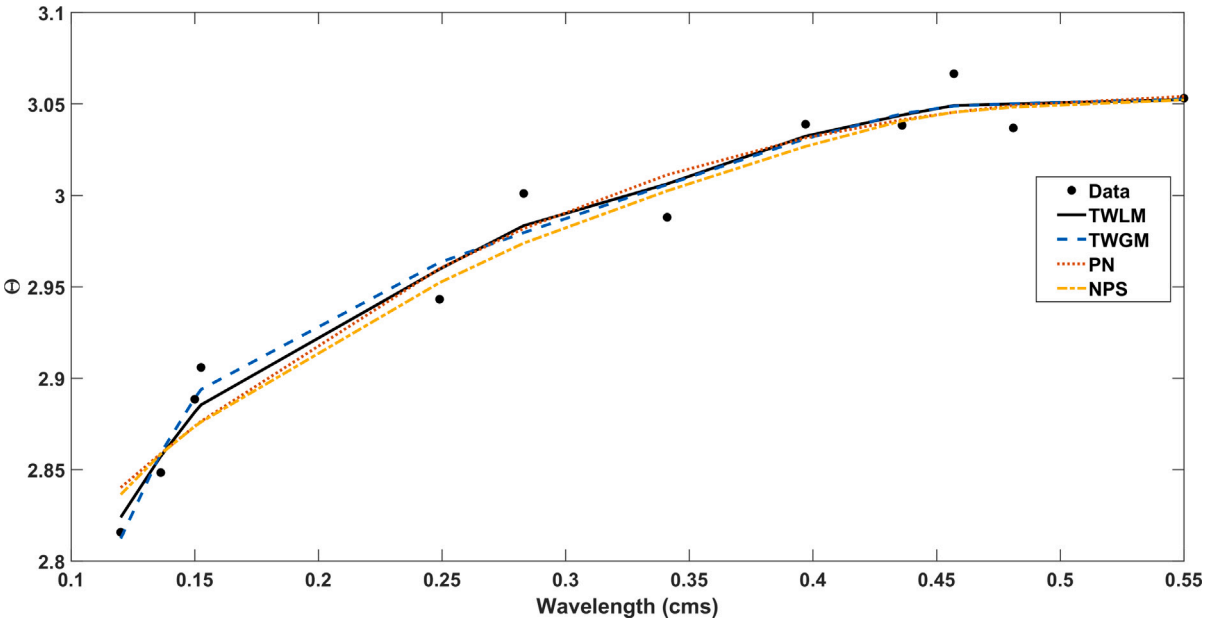


Fig. 5. Estimated mean functions produced by the four models alongside the observed data for the droplet motion application.

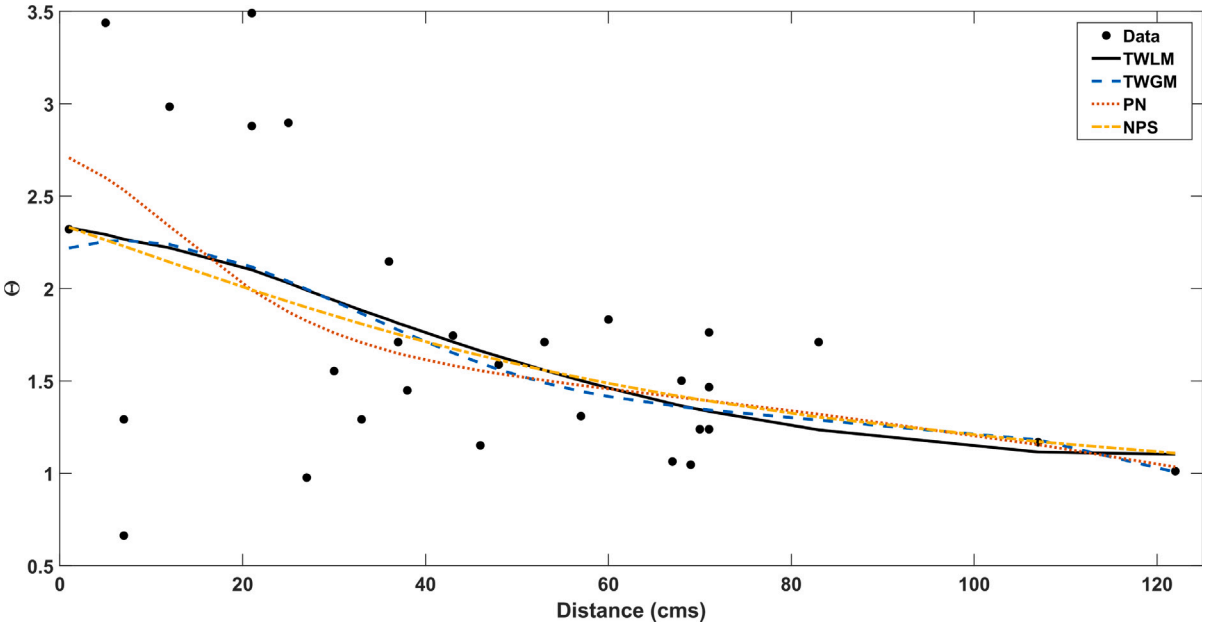


Fig. 6. The estimated mean functions produced by the four models alongside the observed data of blue periwinkles application.

Snail movement prediction

This application analyzes a biological dataset originally provided by Fisher and Lee [36], consisting of movement distances and directions of small blue periwinkles after relocation. The objective is to predict the movement angle based on the distance traveled. This classical dataset serves as a benchmark for circular regression and has been explored in previous studies, including the SPML model by Presnell et al. [43] and the NPS approach by Di Marzio et al. [44].

Table 8 summarizes the predictive performance of the compared methods using MCE, CS, and r_{circ}, along with the computational time required. Fig. 6 illustrates the estimated mean functions overlaid with the observed data.

Table 8
Performance metrics and computation time (in seconds) for movements of blue periwinkles prediction using the four comparative approaches.

Metric	TWLM	TWGM	PN	NPS
MCE	0.2164	0.2198	0.2199	0.2166
CS	0.8414	0.8327	0.8234	0.8315
r_{circ}	0.5594	0.5197	0.4722	0.5077
Time (s)	0.1489	0.0968	6.4275	0.1382

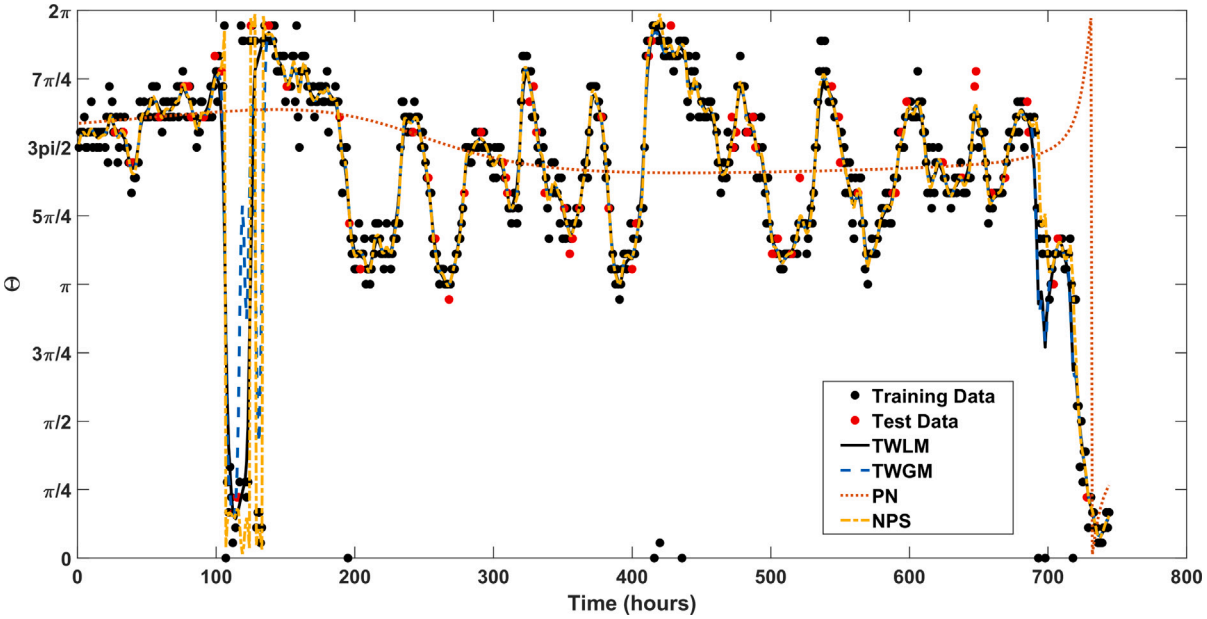


Fig. 7. Estimated mean functions produced by the four models alongside the observed hourly wind direction data.

All four methods yielded comparable MCE values, clustering closely around 0.21. The proposed TWLM achieved the lowest error (0.2164) and the highest Cosine Similarity (0.8414). Notably, TWLM demonstrated superior performance in terms of circular correlation (r_{circ} = 0.5594), significantly outperforming the PN model (0.4722) and the NPS (0.5077). This suggests that TWLM better captures the directional dependence structure of the snail movements.

In terms of computational efficiency, the proposed frameworks (TWGM and TWLM) and the NPS method are extremely fast, with runtimes under 0.15 s. In sharp contrast, the PN model required approximately 6.43 s — over 40 times slower than the proposed methods — likely due to the computational intensity of its Gibbs sampling procedure.

Wind direction prediction

This analysis evaluates the predictive capability of the proposed framework applied to meteorological data, specifically aiming to predict wind direction at unobserved time points based on historical hourly records. The dataset comprises 744 hourly observations collected from January 1 to January 31, 2025, at the Chicago Midway Airport weather station (Source: Midwestern Regional Climate Center, https://mrcc.purdue.edu). The data were partitioned randomly into a training set (90% of observations) and a test set (10%). This random splitting scheme was chosen to evaluate the model's interpolation capabilities specifically, its ability to recover the underlying manifold structure and estimate missing directional values within a known time domain, rather than performing strictly forward-looking forecasting.

Table 9 summarizes the predictive performance using MCE, CS, and r_{circ}, along with the total computation time in seconds. Fig. 7 displays the estimated mean functions overlaid with the observed data, providing a visual assessment of the model fit.

The proposed TWLM and TWGM models demonstrate superior performance, achieving the lowest MCE values (0.0990 and 0.0993, respectively) and the highest correlation metrics. This indicates that the hybrid unfolding strategy effectively captures the rapid directional shifts and wrapping inherent in wind data.

In terms of computational efficiency, both TWLM and TWGM are highly efficient, requiring approximately 2.4 s to train. The standard NPS method remains competitive, trailing closely in accuracy (MCE = 0.1047) with a comparable runtime (2.85 s).

In contrast, the PN model exhibits significant limitations in this application. It yields a substantially higher error (MCE = 0.3465) and lower cosine similarity (0.6309), suggesting an inability to resolve the complex topological structure of the wind time series.

Table 9
Performance metrics and computation time (in seconds) for wind direction prediction using the four comparative approaches.

Metric	TWLM	TWGM	PN	NPS
MCE	0.0990	0.0993	0.3349	0.1047
CS	0.9452	0.9206	0.6309	0.9284
r_{circ}	0.9397	0.9164	0.1715	0.9018
Time (s)	2.45	2.33	32.78	2.85

Furthermore, it is computationally expensive, with a runtime exceeding 32 s—more than an order of magnitude slower than the proposed approach.

Conclusion

In this study, we introduced a novel modeling framework that defined a broad class of nonparametric linear–circular regression models, significantly extending the methodological tools available for circular data analysis. To implement our approach efficiently, we developed an EM algorithm tailored to the structure of our framework. Through extensive numerical experiments and computational evaluations, we compared two special cases of the proposed framework with the PN and NPS models. The results demonstrated that our methods achieved strong performance across a wide range of settings. We further validated the practical utility of the framework using three real-world datasets, highlighting the robustness and adaptability of our models. These findings underscored the flexibility of the proposed framework in capturing complex patterns in circular data and marked a meaningful contribution to the advancement of circular regression modeling.

CRediT authorship contribution statement

E. Zinhom: Conceptualization, Methodology, Software, Writing – original draft. **M.M. Nassar:** Supervision, Project administration, Writing – review & editing. **S.S. Radwan:** Data curation, Formal analysis, Visualization. **A. Elmasry:** Validation, Writing – review & editing.

Ethical approval

Ethical approval is not applicable.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Google Gemini Pro in order to improve the clarity and readability of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data set used is taken from the Midwestern Regional Climate Center (MRCC) https://mrcc.purdue.edu.

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